UNIT-I: DIFFERENTIAL EQUATIONS OF FIRST ORDER & FIRST DEGREE

1. Solve $\frac{dy}{dx} + \frac{y}{x\log x} = \frac{\sin 2x}{\log x}$

Sol. Given equation,

$$\frac{dy}{dx} + \frac{y}{x\log x} = \frac{\sin 2x}{\log x}$$

This is a linear D.E in 'y' with

$$P(x) = \frac{1}{x\log x}, \quad Q(x) = \frac{\sin 2x}{\log x}$$

$$I.F = e^{\int P(x)\,dx} = e^{\int \frac{1}{x\log x}} = e^{\log(\log x)} = \log x$$

G.S is $y(I.F) = \int [Q(x).I.F]\,dx + c$

$$y.\log x = \int \left(\frac{\sin 2x}{\log x}.\log x\right) dx + c$$

$$y.\log x = -\frac{\cos 2x}{2} + c$$

2. $\frac{dy}{dx} + 2y = e^x + x$, $y(0) = 1$

Sol. Given eqn,

$$\frac{dy}{dx} + 2y = e^x + x, \quad y(0) = 1$$

This is a linear D.E in 'y' with $P(x) = 2$,

$$Q(x) = e^x + x$$

$$I.F = e^{\int P(x)\,dx} = e^{\int 2\,dx} = e^{2x}$$

G.S $\Rightarrow (I.F)\,y = \int (Q(x).I.F)\,dx + c$

$$e^{2x}y = \int (e^x + x)(e^{2x})\,dx + c$$

$$e^{2x}y = \int (e^{3x} + xe^{2x})\,dx + c$$

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$$e^{2x}\cdot y = \frac{e^{3x}}{3} + \frac{x e^{2x}}{2} - \frac{e^{2x}}{4} + c$$

$$y = \frac{e^x}{3} + \frac{x}{2} - \frac{1}{4} + c$$

$$y(0) = 1 \Rightarrow 1 = \frac{e^0}{3} + 0 - \frac{1}{4} + c$$

$$1 = \frac{1}{3} - \frac{1}{4} + c$$

$$1 = \frac{1}{12} + c \Rightarrow c = 1 - \frac{1}{12}$$

$$\Rightarrow \boxed{c = \frac{11}{12}}$$

$$y = \frac{e^x}{3} + \frac{x}{2} - \frac{1}{4} + \frac{11}{12}$$

$$y = \frac{e^x}{3} + \frac{x}{2} + \frac{2}{3}$$

3. Solve $dr + (2r\cot\theta + \sin 2\theta)d\theta = 0$

Sol. $dr = -[2r\cot\theta + \sin 2\theta]d\theta$

$$\frac{dr}{d\theta} + 2r\cot\theta = -\sin 2\theta$$

This is linear D.E in 'r', $P(\theta) = 2\cot\theta$,

$Q(\theta) = -\sin 2\theta$

$$I.F = e^{\int P(\theta)d\theta} = e^{\int 2\cot\theta} = e^{2\log(\sin\theta)} = \sin^2\theta$$

G.S $\Rightarrow r.(I.F) = \int (Q(\theta).I.F)d\theta + c$

$$\Rightarrow r.\sin^2\theta = -\int \sin 2\theta . \sin^2\theta \, d\theta + c$$

$$r\sin^2\theta = -\int 2\sin^3\theta\cos\theta \, d\theta + c$$

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$$r\sin^2\theta = -\frac{\sin^4\theta}{2} + c$$

4. $(1+y^2)dx = (\tan^{-1}y - x)\,dy.$

Sol. Given eqn.

$$(1+y^2)dx = (\tan^{-1}y - x)\,dy$$

$$\frac{dx}{dy} = \frac{\tan^{-1}y - x}{1+y^2}$$

$$\frac{dx}{dy} + \frac{x}{1+y^2} = \frac{\tan^{-1}y}{1+y^2}$$

This is linear D.E in 'y' with $P(x) = \frac{1}{1+y^2}$,

$$Q(x) = \frac{\tan^{-1}y}{1+y^2}$$

$$I.F = e^{\int P(x)dx} = e^{\int \frac{1}{1+y^2}dy} = e^{\tan^{-1}y}$$

$$G.S \Rightarrow e^{\tan^{-1}y}\, y = \int \frac{\tan^{-1}y}{1+y^2} e^{\tan^{-1}y} + c$$

put $\tan^{-1}y = t$

$$\frac{1}{1+y^2}dy = dt \Rightarrow dy = dt(1+y^2)$$

$$\Rightarrow e^t y = \int \frac{t}{1+y^2} e^t (1+y^2)dt + c$$

$$e^t y = \int t\, e^t\, dt + c$$

$$e^t y = e^t(t-1) + c$$

$$y = (t-1) + ce^{-t} \Rightarrow y = \tan^{-1}y - 1 + ce^{-\tan^{-1}y}$$

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5. Solve $(2xy+y-\tan y)dx + (x^2 - x\tan^2 y + \sec^2 y)dy$

Sol. Given eqn,

$(2xy + y - \tan y)dx + (x^2 - x\tan^2 y + \sec^2 y)dy = 0$

Which is in the form of $Mdx + Ndy = 0$

$M = 2xy + y - \tan y$, $N = x^2 - x\tan^2 y + \sec^2 y$

$\frac{\partial M}{\partial y} = 2x + 1 - \sec^2 y$ $\quad \frac{\partial N}{\partial x} = 2x - \tan^2 y$

$= 2x + 1 - 1 - \tan^2 y$

$= 2x - \tan^2 y$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ i.e. given eqn is exact

G.S is $\int Mdx + \int Ndy = c$

$\int 2xy + y - \tan y\, dx + \int \sec^2 y\, dy = c$

$x^2y + xy - x\tan y + \tan y = c$

6. $(y^2 e^{xy^2} + 4x^3)dx + (2xye^{xy^2} - 3y^2)dy = 0$

Sol. Given eqn.

$(y^2 e^{xy^2} + 4x^3)dx + (2xye^{xy^2} - 3y^2)dy = 0$

Which is in the form of $Mdx + Ndy = 0$

$M = y^2 e^{xy^2} + 4x^3$, $N = 2xye^{xy^2} - 3y^2$

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$\frac{\partial M}{\partial y} = 2ye^{xy^2} + 2y^3xe^{xy^2}$ $\frac{\partial N}{\partial x} = 2ye^{xy^2} + 2xy^3e^{xy^2}$

$= 2e^{xy^2}y(xy^2+1)$ $\frac{\partial N}{\partial x} = 2ye^{xy^2}[1+xy^2]$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ i.e. given eqn is exact

G.S is $\int M\,dx + \int N\,dy = C$

$\int 2ye^{xy^2} + x$

$\int (y^2e^{xy^2} + 4x^3)\,dx + \int -3y^2\,dy = C$

$\frac{y^2e^{xy^2}}{y^2} + \frac{4x^4}{4} - \frac{3y^3}{3} = C$

$e^{xy^2} + x^4 - y^3 = C$

7. solve $\frac{dy}{dx} + \frac{y\cos x + \sin y + y}{\sin x + x\cos y + x} = 0$

Sol) Given eqn

$\frac{dy}{dx} + \frac{y\cos x + \sin y + y}{\sin x + x\cos y + x} = 0$

$(\sin x + x\cos y + x)\,dy + (y\cos x + \sin y + y)\,dx = 0$

Which is in the form of $M\,dx + N\,dy = 0$

$M = y\cos x + \sin y + y$ $N = \sin x + x\cos y + x$

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$\frac{\partial M}{\partial y} = \cos x + \cos y + 1$ $\quad \frac{\partial N}{\partial x} = \cos x + \cos y + 1$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ i.e. given eqn is exact.

G.S $\Rightarrow \int M dx + \int N dy = c$

$\int y\cos x + \sin y + y + \int 0 \, dy = c$

$y\sin x + (\sin y + y)x = c$

8. Solve $(x^2y - 2xy^2)dx + (x^3 - 3x^2y)dy = 0$

Sol. Given eqn

$(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0$

which is in the form of $Mdx + Ndy = 0$

$M = x^2y - 2xy^2, \quad N = -x^3 + 3x^2y$

$\frac{\partial M}{\partial y} = x^2 - 4x \quad \frac{\partial N}{\partial x} = -3x^2 + 6xy$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ i.e. given eqn is not exact

Since eqn is not exact & homogeneous

$Mx + Ny \Rightarrow x^3y - 2x^2y^2 - x^3y + 3x^2y^2$

$\Rightarrow x^2y^2 \neq 0$

$I.F = \frac{1}{Mx + Ny} = \frac{1}{x^2y^2}$

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Multiply the eqn with I.F.

$$\frac{1}{x^2y^2}[x^2y - 2xy^2]\,dx + \frac{1}{x^2y^2}[-x^3 + 3x^2y]\,dy = 0$$

$$\left[\frac{1}{y} - \frac{2}{x}\right]dx + \left[-\frac{x}{y^2} + \frac{3}{y}\right]dy = 0$$

which is in the form of $M_1\,dx + N_1\,dy = 0$

$$M_1 = \frac{1}{y} - \frac{2}{x}, \quad N_1 = \frac{3}{y} - \frac{x}{y^2}$$

$$\frac{\partial M_1}{\partial y} = -\frac{1}{y^2} \qquad \frac{\partial N_1}{\partial x} = -\frac{1}{y^2}$$

$\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$ i.e. given eqn is exact

G.S $\Rightarrow \int M_1\,dx + \int N_1\,dy = C$

$$\int\left(\frac{1}{y} - \frac{2}{x}\right)dx + \int\left(\frac{3}{y} - \frac{x}{y^2}\right)dy = C$$

$$\frac{x}{y} - 2\log x + 3\log y = C$$

$$\frac{x}{y} + \log x^{-2} + \log y^3 = C$$

$$\frac{x}{y} + \log\left(y^3/x^2\right) = C$$

9. $y(xy\sin xy + \cos xy)\,dx + x(xy\sin xy - \cos xy)\,dy =$

Sol. Given eqn

$y(xy\sin xy + \cos xy)\,dx + x(xy\sin xy - \cos xy)\,dy = 0$

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$M = xy^2\sin xy + y\cos xy$ $\quad N = x^2y\sin xy - x\cos xy$

$$\frac{\partial M}{\partial y} = 2xy\sin xy + x^2y^2\cos xy + \cos xy + xy\sin xy$$

$$= xy\sin xy + \cos xy[x^2y^2 + 1]$$

$$\frac{\partial N}{\partial x} = 2xy\sin xy + x^2y^2\cos xy - \cos xy + xy\sin xy$$

$$= 3xy\sin xy + \cos xy[x^2y^2 - 1]$$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ i.e. given eqn is not exact.

& it is in the form of $\int f(x,y)\,dx + \int g(x,y)\,dy = c$

& $Mx - Ny$ =)

$$x^2y^2\sin xy + xy\cos xy - x^2y^2\sin xy + xy\cos xy$$

=) $2xy\cos xy \neq 0$

$$I.F = \frac{1}{Mx - Ny} = \frac{1}{2xy\cos xy}$$

Multiply the eqn with I.F.

$$\frac{}{2xy\cos xy}[xy\sin xy + \cos xy]\,dx + \frac{x}{2xy\cos xy}[xy\sin xy - \cos xy]\,dy = 0$$

$$\left[\frac{y}{2}\tan xy + \frac{1}{2x}\right]dx + \left[\frac{x}{2}\tan xy - \frac{1}{2y}\right]dy = 0$$

which is in the form of $M_1\,dx + N_1\,dy = 0$

$\frac{\partial M_1}{\partial y} = \frac{1}{2}\tan xy + \frac{xy}{2}\sec^2 xy$

$\frac{\partial N_1}{\partial x} = \frac{1}{2}\tan xy + \frac{xy}{2}\sec^2 xy$

$\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$ ie. given eqn is exact.

G.S $\Rightarrow \int M_1\,dx + \int N_1\,dy = c$

$$\int\left[\frac{y}{2}\tan xy + \frac{1}{2x}\right]dx + \int -\frac{1}{2y}\,dy = c$$

$$\frac{y}{2}\,\frac{\log(\sec xy)}{y} + \frac{1}{2}\log x - \frac{1}{2}\log y = c$$

$$\frac{1}{2}\left[\log(\sec xy)\left(\frac{x}{y}\right)\right] = c$$

10. $\frac{dy}{dx} = \frac{x^2+y^2+1}{2xy}$

Sol. Given eqn.

$\frac{dy}{dx} = \frac{x^2+y^2+1}{2xy}$

$(x^2+y^2+1)dx - 2xy\,dy = 0$

which is in the form of $M dx + N dy = 0$

$M = x^2+y^2+1$, $N = -2xy$

$\frac{\partial M}{\partial y} = 2y$ $\quad \frac{\partial N}{\partial x} = -2y.$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ i.e. given eqn is not exact.

$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} / N \Rightarrow \frac{2y+2y}{-2xy} = -\frac{2}{x}$

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I.F $\Rightarrow e^{-\int \frac{2}{x} dx} \Rightarrow e^{-2\log x} = x^{-2} = \frac{1}{x^2}$

$$\frac{1}{x^2}\left[x^2+y^2+1\right]dx - \frac{2xy}{x^2}dy = 0$$

$$\left[1+\frac{y^2}{x^2}+\frac{1}{x^2}\right]dx - \frac{2y}{x}dy = 0$$

which is in the form of $M_1 dx + N_1 dy = 0$

$$M_1 = 1+\frac{y^2}{x^2}+\frac{1}{x^2} \qquad N_1 = -\frac{2y}{x}$$

$$\frac{\partial M_1}{\partial y} = \frac{2y}{x^2} \qquad \frac{\partial N_1}{\partial x} = +\frac{2y}{x^2}$$

$\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$ i.e. given eqn is exact.

G.S $\Rightarrow \int M_1 dx + \int N_1 dy = c$

$$\int\left[1+\frac{y^2}{x^2}+\frac{1}{x^2}\right]dx + \int 0\, dy = c$$

$$x - \frac{y^2}{x} - \frac{1}{x} = c$$

11. Solve $(xy^3+y)dx + 2(x^2y^2+x+y^4)dy = 0$.

Sol) Given eqn.

$$(xy^3+y)dx + 2(x^2y^2+x+y^4)dy = 0$$

which is in the form of $Mdx + Ndy = 0$

$$M = xy^3+y. \qquad N = 2x^2y^2+2x+2y^4$$

$$\frac{\partial M}{\partial y} = 3xy^2+1 \qquad \frac{\partial N}{\partial x} = 4xy^2+2$$

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$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ i.e. given eqn is not exact

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} \Rightarrow \frac{4xy^2 + 2 - 3xy^2 - 1}{y(xy^2+1)}$$

$$\Rightarrow \frac{xy^2+1}{y(xy^2+1)} = \frac{1}{y}$$

$$I.F = e^{\int \frac{1}{y} dy} = e^{\log y} = y$$

Multiply the eqn with I.F.

$$y(xy^3+y)dx + (2x^2y^2+2x+2y^4)y\,dy = 0$$

$$(xy^4+y^2)dx + (2x^2y^3+2xy+2y^5)dy = 0$$

Which is in the form of $M_1dx + N_1dy = 0$

$M_1 = xy^4+y^2$ $\quad N_1 = 2x^2y^3+2xy+2y^5$

$\frac{\partial M_1}{\partial y} = 4xy^3+2y$ $\quad \frac{\partial N_1}{\partial x} = 4xy^3+2y$

$\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$ i.e. given eqn is exact

G.S $\Rightarrow \int M_1 dx + \int N_1 dy = C$

$$\int xy^4+y^2 dx + \int 2y^5\,dy = C$$

$$\frac{x^2y^4}{2} + xy^2 + \frac{2y^6}{6} = C$$

$$\frac{x^2y^4}{2} + xy^2 + \frac{y^6}{3} = C$$

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12. $x^4 \frac{dy}{dx} + x^3y + \text{cosec}(xy) = 0$

Sol. Given eqn.

$$x^4 \frac{dy}{dx} + x^3y + \text{cosec}(xy) = 0$$

$$x^4 dy + [x^3y + \text{cosec}(xy)]dx = 0$$

which is in the form of $M dx + N dy = 0$

$M = x^3y + \text{cosec}(xy)$ $\quad N = x^4$

$\frac{\partial M}{\partial y} = x^3 + \text{cosec}(xy)\cot(xy)$ $\quad \frac{\partial N}{\partial x} = 4x^3$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ i.e. given eqn not exact

Divide the whole eqn with x^4

$$\frac{dy}{dx} + \frac{y}{x} = -\frac{\text{cosec}(xy)}{x^4}$$

$xy = t$

$$\left[\frac{dt}{dx} - y\right]\frac{1}{x} + \frac{y}{x} = -\frac{\text{cosec}\,t}{x^4}$$

$$xy + x\frac{dy}{dx} = \frac{dt}{dx}$$

$$\left[\frac{dt}{dx}\right]\frac{1}{x} = -\frac{\text{cosec}\,t}{x^4}$$

$$\frac{dy}{dx} = \left[\frac{dt}{dx} - y\right]\frac{1}{x}$$

$$\frac{dt}{dx} = -\frac{\text{cosec}\,t}{x^3}$$

$$\frac{dt}{dx} = -\frac{1}{\sin t \cdot x^3}$$

$$+\sin t\,dt = \frac{1}{x^3}\,dx$$

$$\int \sin t\,dt = -\int \frac{1}{x^3}\,dx$$

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$$-\cos t = \frac{-[\quad]}{-2x^2}\,dx + c$$

$$-\cos t = \frac{1}{2x^2} + c$$

$$\cos t = -\frac{1}{2x^2} + c$$

$$\cos(xy) = -\frac{1}{2x^2} + c$$

13. Find the orthogonal trajectories of family of circles $x^2+y^2+2\lambda x+c=0$ where 'λ' is parameter.

Sol. Given family,

$$x^2+y^2+2\lambda x+c=0$$

diff. w.r.t. 'x'

$$2x + 2y\frac{dy}{dx} + 2\lambda = 0$$

$$x + y\frac{dy}{dx} + \lambda = 0$$

$$\lambda = -\left[x + y\frac{dy}{dx}\right]$$

$$\Rightarrow x^2+y^2+2\left[-x-y\frac{dy}{dx}\right]x + c = 0$$

$$x^2+y^2-2x^2-2xy\frac{dy}{dx} + c = 0$$

$$y^2-x^2-2xy\frac{dy}{dx} + c = 0 \Rightarrow$$ This is D.E of given family

Replace $\frac{dy}{dx}$ with $\frac{-dx}{dy}$

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$y^2 - x^2 + 2xy\frac{dx}{dy} + c = 0$

$\Rightarrow (y^2 - x^2 + c) + 2xy\frac{dx}{dy} = 0$

$(y^2 - x^2 + c)dy + 2xy\,dx = 0$

This is D.E of orthogonal family.

$(y^2 - x^2 + c)dy + 2xy\,dx = 0$

This is in the form of $M dx + N dy = 0$

$M = 2xy \qquad N = y^2 - x^2 + c$

$\frac{\partial M}{\partial y} = 2x \qquad \frac{\partial N}{\partial x} = -2x$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ i.e. given eqn is not exact

$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{M} \Rightarrow \frac{-2x - 2x}{2xy} = \frac{-4x}{2xy} = -\frac{2}{y}$

$I.F = e^{\int -\frac{2}{y}dy} = e^{-2\log y} = \frac{1}{y^2}$

$\frac{1}{y^2}(y^2 - x^2 + c)dy + \frac{1}{y^2}(2xy)dx = 0$

$\left[1 - \frac{x^2}{y^2} + \frac{c}{y^2}\right]dy + \frac{2x}{y}dx = 0$

This is in the form of $M_1 dx + N_1 dy = 0$

$M_1 = \frac{2x}{y}, \qquad N_1 = 1 - \frac{x^2}{y^2} + \frac{c}{y^2}$

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$\frac{\partial M_1}{\partial y} = \frac{-2x}{y^2}$ $\quad \frac{\partial N_1}{\partial x} = \frac{+2x}{y^2}$

$\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$ i.e. given eqn is exact.

G.S $\Rightarrow$ $\int M dx + \int (\text{terms of } N \text{ free of } x)\, dy = k$

$$\int \frac{2x}{y} dx + \int \frac{c}{y^2} dy = k$$

$$\frac{x^2}{y} + \frac{c}{y} + y = k$$

$$\frac{y^2 + x^2 - c}{y} = k$$

14. Show that the following curves are self orthogonal curves

(i) $y^2 = 4a(x+a)$, a is parameter

(ii) $\frac{x^2}{a^2+\lambda} + \frac{y^2}{b^2+\lambda} = 1$, λ is parameter.

Sol. (i) Given family,

$$y^2 = 4a(x+a)$$

diff. w.r.t. 'x'

$$2y \frac{dy}{dx} = 4a$$

$$a = \frac{y}{2} \frac{dy}{dx}$$

$$\Rightarrow y^2 = 4\left[\frac{y}{2} \frac{dy}{dx}\right]\left[x + \frac{y}{2} \frac{dy}{dx}\right]$$

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$$y^2 = 2xy\frac{dy}{dx} + \frac{y^2}{2}\frac{dy}{dx}\cdot\frac{dy}{dx}$$

$$y^2 = \frac{dy}{dx}\left[2xy + \frac{y^2}{2}\frac{dy}{dx}\right]$$

$$y = \frac{dy}{dx}\left[2x + y\frac{dy}{dx}\right] \rightarrow$$ This is D.E of given family.

Now replace $\frac{dy}{dx}$ with $-\frac{dx}{dy}$

$$y = -\frac{dx}{dy}\left[2x - y\frac{dx}{dy}\right]$$

$$\frac{dy}{dx}\cdot y = -2x + y\frac{dx}{dy}$$

$$\frac{dy}{dx}\cdot y\cdot\frac{dy}{dx} = -2x\frac{dy}{dx} + y$$

$$\left[y\frac{dy}{dx}\right]\frac{dy}{dx} + 2x\frac{dy}{dx} = y$$

$$y = \left[2x + y\frac{dy}{dx}\right]\frac{dy}{dx}$$

This is D.E of orthogonal family.

Since, the D.Eq^{n}s of given family and orthogonal family are equal, the given family is self orthogonal.

(ii) Given family

$$\frac{x^2}{a^2+\lambda^2} + \frac{y^2}{b^2+\lambda^2} = 1$$

diff. wrt. x.

$$\frac{2x}{a^2+\lambda} + \frac{2yy'}{b^2+\lambda} = 0$$

$$(b^2+\lambda)2x + (2yy')(a^2+\lambda) = 0$$

~~$\lambda(2x + 2yy' \lambda(a^2$~~

$$(x+yy')\lambda + (a^2yy' + b^2x) = 0$$

$$\lambda = -\frac{(a^2yy' + b^2x)}{x+yy'}$$

$$\Rightarrow a^2+\lambda = a^2 - \left[\frac{a^2yy' + b^2x}{x+yy'}\right]$$

$$= \frac{a^2(x+yy') - [a^2yy' + b^2x]}{x^2+yy'}$$

$$= \frac{a^2x - b^2x}{x+yy'} = \frac{x(a^2-b^2)}{x+yy'}$$

$$\Rightarrow b^2+\lambda = b^2 - \left[\frac{a^2yy' + b^2x}{x+yy'}\right]$$

$$= \frac{b^2(x+yy') - a^2yy' - b^2x}{x+yy'}$$

$$= \frac{yy'(b^2-a^2)}{x+yy'}$$

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$$\Rightarrow \frac{x^2}{\frac{x(a^2-b^2)}{x+yy'}} + \frac{y^2}{\frac{yy'(b^2-a^2)}{x+yy'}} = 1$$

$$\frac{x(x+yy')}{a^2-b^2} + \frac{y(x+yy')}{y'(b^2-a^2)} = 1$$

$$\frac{x+yy'}{a^2-b^2}\left[x - \frac{y}{y'}\right] = 1$$

This is D.E of given family.

Now, replace y' with $-\frac{1}{y'}$

$$\frac{x+y\left[-\frac{1}{y'}\right]}{a^2-b^2}\left[x + yy'\right] = 1$$

$$\frac{xy'-y}{y'(a^2-b^2)}\left[x + yy'\right] = 1$$

$$\frac{(x+yy')}{a^2-b^2}\frac{[x - y/y']}{} = 1$$

This is DE of orthogonal family.

∴ The D.E eqns of both families are equal.

Given family is self orthogonal.

15. Find the orthogonal trajectories of the following curves.

(i) $r = a(1-\cos\theta)$ (ii) $r = 2a(\cos\theta + \sin\theta)$

(i) $r = a(1-\cos\theta)$

diff- w.r.t 'θ'

$$\frac{dr}{d\theta} = a\sin\theta$$

$$a = \frac{dr}{d\theta}\frac{1}{\sin\theta}$$

$$\Rightarrow r = \frac{\frac{dr}{d\theta}}{\sin\theta}[1-\cos\theta]$$

$$r\sin\theta = \frac{dr}{d\theta}[1-\cos\theta]$$

$$\frac{dr}{d\theta} = \frac{r\sin\theta}{1-\cos\theta}$$

$$\frac{dr}{d\theta} = \frac{r\,2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}}$$

$$\frac{dr}{d\theta} = r\cot\frac{\theta}{2}$$

Now replace $\frac{dr}{d\theta}$ with $-r^2\frac{d\theta}{dr}$

$$-r^2\frac{d\theta}{dr} = r\cot\frac{\theta}{2}$$

$$-r\frac{d\theta}{dr} = \cot\frac{\theta}{2}$$

$$-\tan\frac{\theta}{2}\,d\theta = \frac{1}{r}\,dr$$

Integrating on both sides

$$\frac{-\log \sec\left(\frac{\theta}{2}\right)}{\frac{1}{2}} + \log c = \log r$$

$$\log r = \log\left(\cos^2\frac{\theta}{2}\right) + \log 2c$$

$$\log r = \log\left[\cos^2\frac{\theta}{2}\right] 2c$$

$$r = 2c\left[\frac{1+\cos\theta}{2}\right]$$

$$r = c[1+\cos\theta]$$

(ii) Given family,

$$r = 2a(\cos\theta + \sin\theta)$$

diff. w.r.t. θ.

$$\frac{dr}{d\theta} = -2a\sin\theta + 2a\cos\theta$$

$$\frac{dr}{d\theta} = 2a(\cos\theta - \sin\theta)$$

$$2a = \frac{dr}{d\theta}\left[\frac{1}{\cos\theta - \sin\theta}\right]$$

$$\Rightarrow r = \frac{dr}{d\theta}\left[\frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta}\right]$$

$$\left[\frac{\cos\theta - \sin\theta}{\cos\theta + \sin\theta}\right] d\theta = \frac{1}{r} dr \rightarrow \left[\frac{\cos\theta - \sin\theta}{\cos\theta + \sin\theta}\right] r = \frac{dr}{d\theta}$$

replace $\frac{dr}{d\theta}$ with $-r^2\frac{d\theta}{dr}$

$$-r^2 \frac{d\theta}{dr} = r\left[\frac{\cos\theta - \sin\theta}{\cos\theta + \sin\theta}\right]$$

$$-r\frac{d\theta}{dr} = \frac{\cos\theta - \sin\theta}{\cos\theta + \sin\theta}$$

$$\left[\frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta}\right] d\theta = -\frac{1}{r}\, dr$$

Integrating on both sides

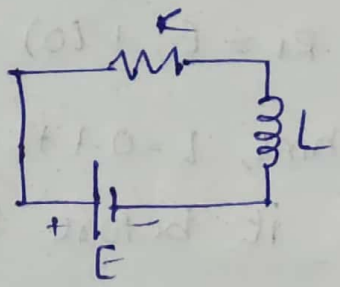
$$-\int \frac{-(\cos\theta + \sin\theta)}{\cos\theta - \sin\theta}\, d\theta = -\int \frac{1}{r}\, dr$$

$$-\log(\cos\theta - \sin\theta) = -\log r + \log c$$

$$\cos\theta - \sin\theta = rc$$

16. Model R-L circuit, write its D.E & find the current i at any time 't' in the circuit

Sol) Consider, a circuit with resistance 'R', Inductance 'L' & emf 'E' connected in series.



The voltage drop across resistance = Ri

The voltage drop across Inductance = $L\frac{di}{dt}$

By Kirchoff's law,

$$Ri + L\frac{di}{dt} = E$$

$$\frac{di}{dt} + \frac{R}{L}i = \frac{E}{L}$$

This is linear d.E in 'i'.

with $P(t) = \frac{R}{L}$, $Q(t) = \frac{E}{L}$

$$I.F = e^{\int \frac{R}{L} dt} = e^{Rt/L}$$

G.S $\Rightarrow$ $i.[I.F] = \int [Q(t).I.F]\,dt + c$

$$i.e^{Rt/L} = \int \frac{E}{L} \cdot e^{Rt/L} dt + c$$

$$i.e^{Rt/L} = \frac{\frac{E}{L} e^{Rt/L}}{\frac{R}{L}} + c$$

$$i = \frac{E}{R} + ce^{-Rt/L}$$

17. Find the current in electrical circuit given by $L\frac{di}{dt} + Ri = E$, $i(0) = 0$ where $E = 6$ volts. $R = 100$ ohms, $L = 0.1$ H. Also find how long will it before the current has reached one half of its max.

Sol: Given,

$E = 6V$

$R = 100\ \Omega$

$L = 0.1\ H$

$$1000i + 0.1\frac{di}{dt} = 6$$

$$\frac{di}{dt} + 1000i = 60$$

This is Linear D.E in 'i' with

$$P(t) = 1000, \quad Q(t) = 60$$

$$I.F = e^{\int 1000\,dt} = e^{1000t}$$

G.S $\Rightarrow$ $i.(I.F) = \int (Q(t) \cdot I.F)\,dt + C$

$$i.(e^{1000t}) = \left[\frac{e^{1000t}}{1000} \cdot 60\right] + C$$

$$i = \frac{6}{100} + Ce^{-1000t}$$

initially $i = 0$ at $t = 0$

$$0 = \frac{6}{100} + Ce^{(0)}$$

$$C = -\frac{6}{100}$$

$$\Rightarrow i = \frac{6}{100} - \frac{6}{100}e^{-1000t}$$

$$i = \frac{6}{100}[1 - e^{-1000t}] \quad -(1)$$

By the eqn $i_{max} = \frac{6}{100}$

To find 't' at half of the imax.

$$i = \frac{1}{2}[i_{max}]$$

$$\frac{6}{100}[1 - e^{-1000t}] = \frac{1}{2}\left[\frac{6}{100}\right]$$

$$1 - e^{-1000t} = \frac{1}{2}$$

$$\frac{1}{2} = e^{-1000t}$$

$$-\log 2 = -1000t$$

$$\log 2 = 1000t$$

$$t = \frac{\log 2}{1000}$$

$$t = 0.000301$$